1. **(12 points)** The circuit is shown in Figure 1. 12

(1) Use mesh analysis to obtain u_1.

(2) Calculate the power absorbed by the voltage source.

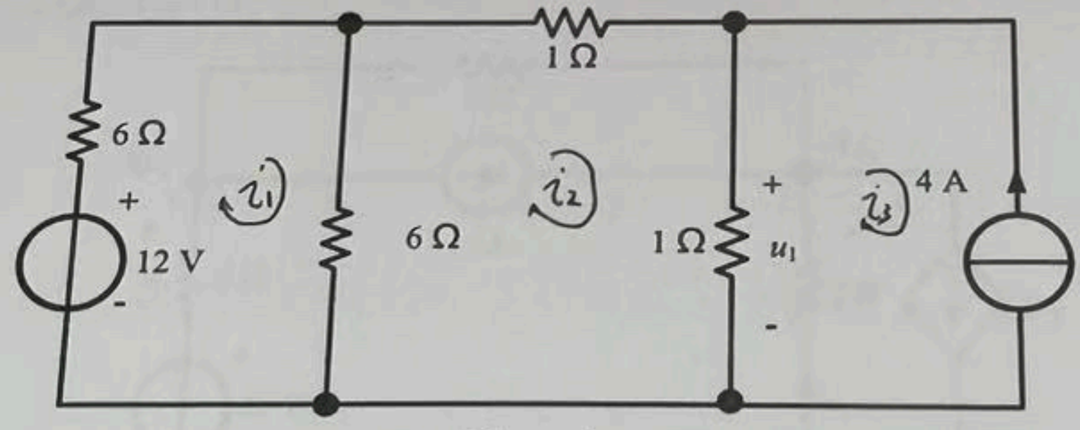


Figure 1

2. **(14 points)** The circuit is shown in Figure 2. 14

(1) Use superposition to obtain i_1.

(2) Employ node analysis to determine i_1 and compare the outcome with that obtained in (1).

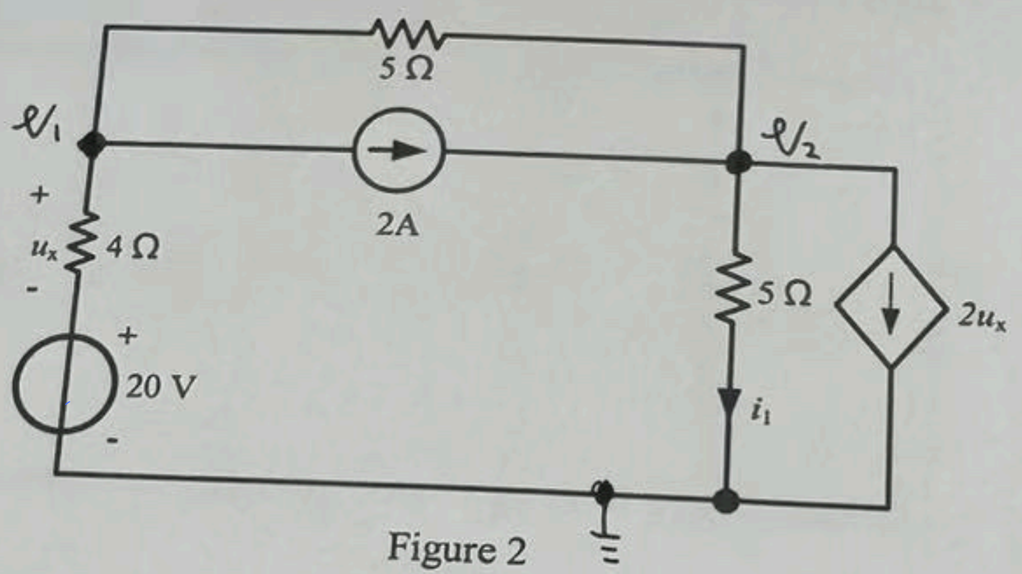


Figure 2

5Ω

3. **(16 points)** The circuit is shown in Figure 3. 10

(1) Determine the Thevenin equivalent with respect to terminals a and b for the circuit, excluding the variable resistor Rx.

(2) Obtain the maximum power consumed by the variable resistor Rx.

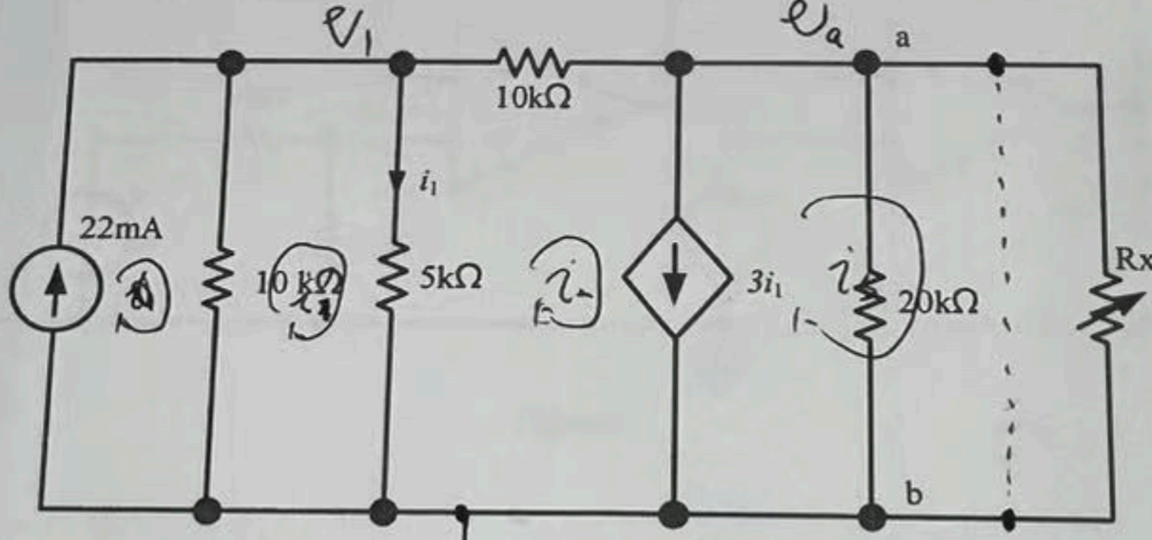


Figure 3

12 4. **(12 points)** The operational amplifiers in the circuit are ideal and operate in their linear range. Obtain u_o in the circuit.

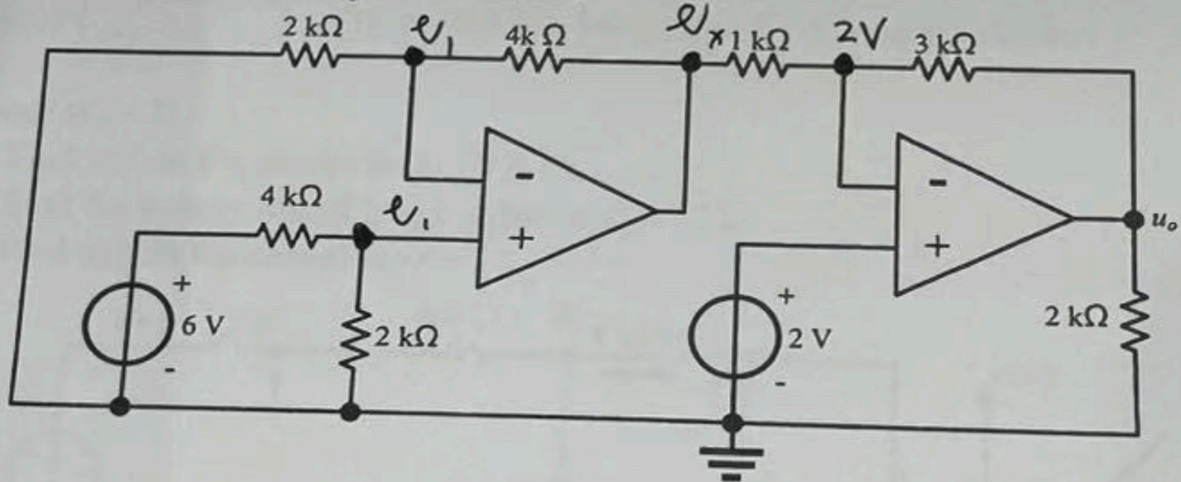


Figure 4

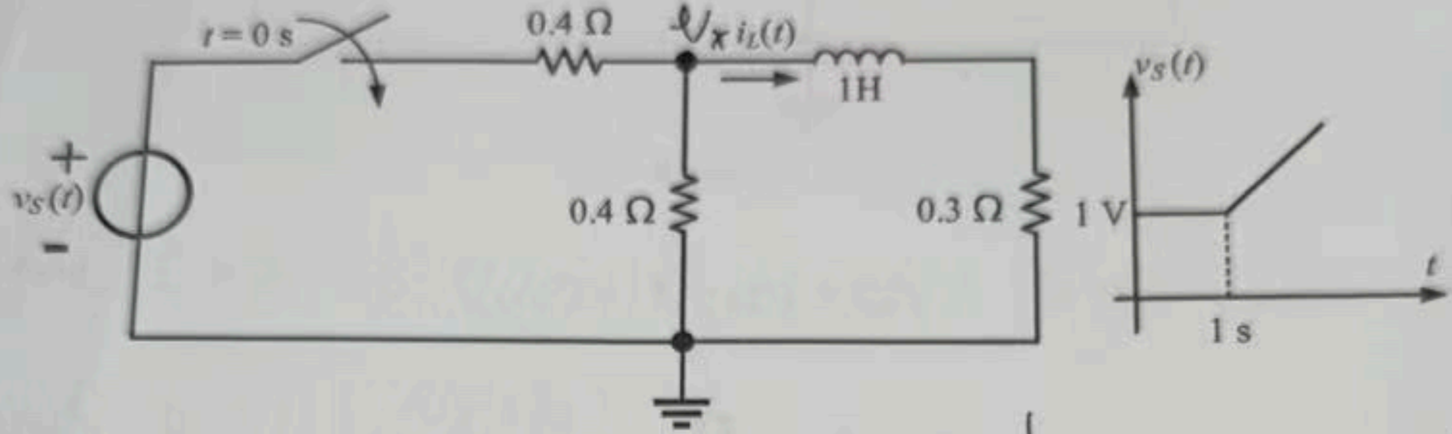
16

5. **(16 points)** The circuit is shown in Figure 5. The voltage source $v_S(t)$ satisfies $v_S(t)=\begin{cases}1\text{ V}, & 0\le t\le 1\text{ s}\\ t\text{ V}, & t>1\text{ s}\end{cases}$. The switch has been open for a long time before t = 0, and is closed at $t=0$.

(1) Find $i_L(t)$ in the circuit for $0\le t\le 1$ s.

(2) Find the energy stored in the inductor at $t=1$ s.

(3) Find $i_L(t)$ in the circuit for $t>1$ s.



14

6. **(14 points)** The circuit is shown in Figure 6. At $t=0$-, $v_C(0\text{-})=0$ V and $i_L(0\text{-})=0$ A. Find $v_C(t)$ in the circuit ($t>0$).

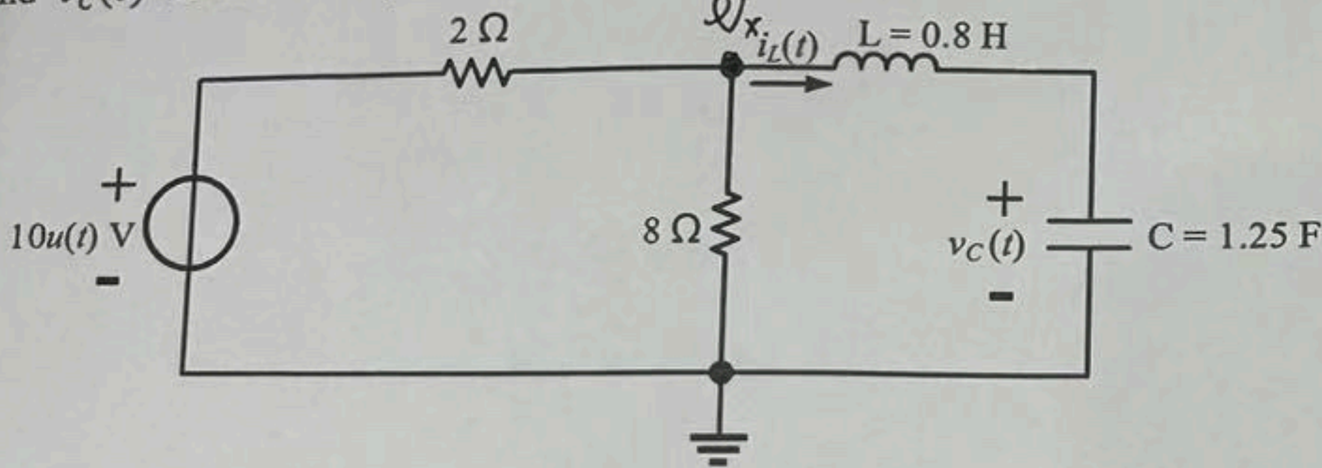


Figure 6

16

7. **(16 points)** The circuit is shown in Figure 7. $v_S(t) = e^{-0.5t}$ V. At $t = 0$-, $v_{C1}(0\text{-}) = 0$V and $v_{C2}(0\text{-}) = 0$V. Find $v_{C2}(t)$ in the circuit ($t > 0$).

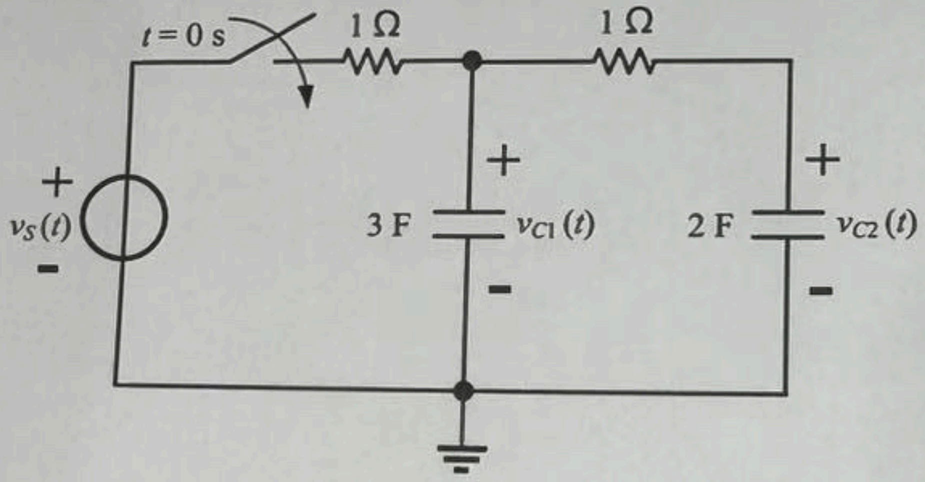


Figure 7